

Projects in the Machine Learning Lab:

Imitation Learning:

Using a camera a neural network is trained. A task is performed within the defined environment, and the model learns the task. The goal is basically a situation where one robot arm is worked on, and the other must mirror its movements by taking the inputs from the cameras and the motors used in the robotic arm.

Zauberzeug Path Finder:

Zauberzeug is a path finder robot focused on path recognition using navigation and mapping techniques.

A Humanoid Robot:

The Humanoid Robot has an upper body as the physical representation. The arms have muscles which seem to be 3D printed. The robot is programmed to engage particularly in playing the piano.

Turtle Robots:

The Turtle Robots are close to five in number and enclosed within a defined environment. They are programmed and tested within this environment and are usually used by students of the bachelor's programme in Robotics.

Project VineBot:

A robot for viticulture programmed for automating tasks in the vineyard.

Application scenarios:

- Controlling the growth of unwanted plants.
- Identification of diseases in grapevines at an early stage.
- Growth monitoring and surveillance.

Research topics:

- Autonomous navigation.

- Image classification and machine learning for weed removal and crop monitoring.
- Manipulators for pruning and harvesting.

Built-in Sensors:

- LIDAR: Livox Mid 40 (Range: 260m, FOV: 38.4°)
- 3D Camera: Intel RealSense D455 (Range: 0.4–6m)
- IMUs: 2 × BNO055
- Rotary encoders in the motors

Mechanical Properties:

- Weight: 140 kg
- Battery: 4 × 12V DC LEAD-ACID 20AH
- Max speed: 3 m/s
- Max torque: 40 Nm
- Max slope: 45°
- Watts: 960 Wh

3D Scan of the lab.